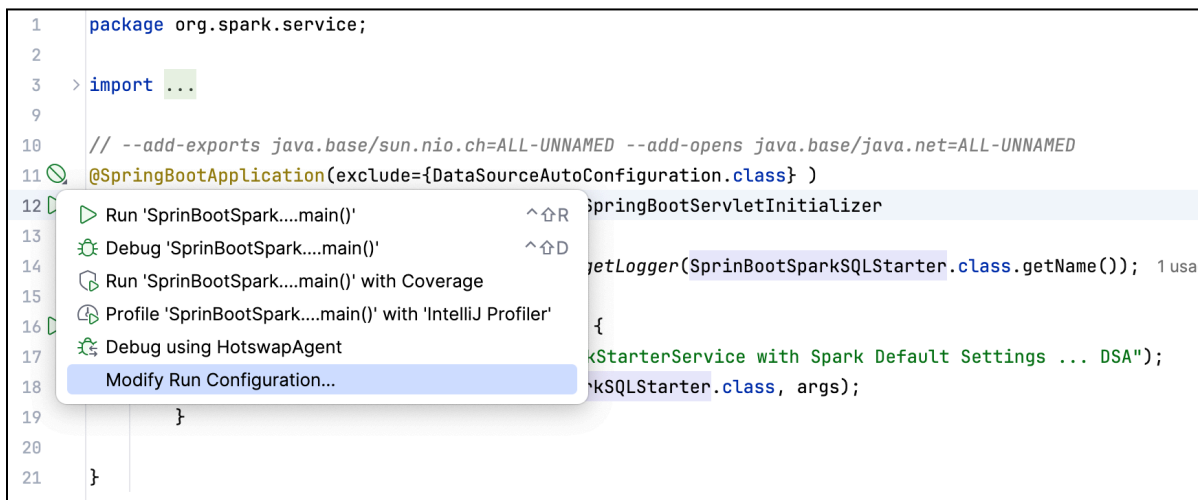


PREPARE SparkSQL Server in LOCAL mode

- **Download** SparkSQL Server as Spring Boot Service:
 - **[DSA-SparkSQL-Service]** Project (maven-based).
- **RUN SparkSQL** in **LOCAL mode** as a Spring Boot application:
 - Prepare SpringBoot-SparkSQL Run Configuration - fix for SparkSQL libraries' *incompatibilities* with JDK 21 (JDK version superior to JDK 11):
 - Open Java Project Module [DSA-SparkSQL-Service].
 - Open SBT application class ***SprinBootSparkSQLStarter*** from
 - package: org.spark.service.
 - Open *SprinBootSparkSQLStarter* Run Configuration:
 - Click on [Run]button alongside the class name.



The screenshot shows a code editor with the following content:

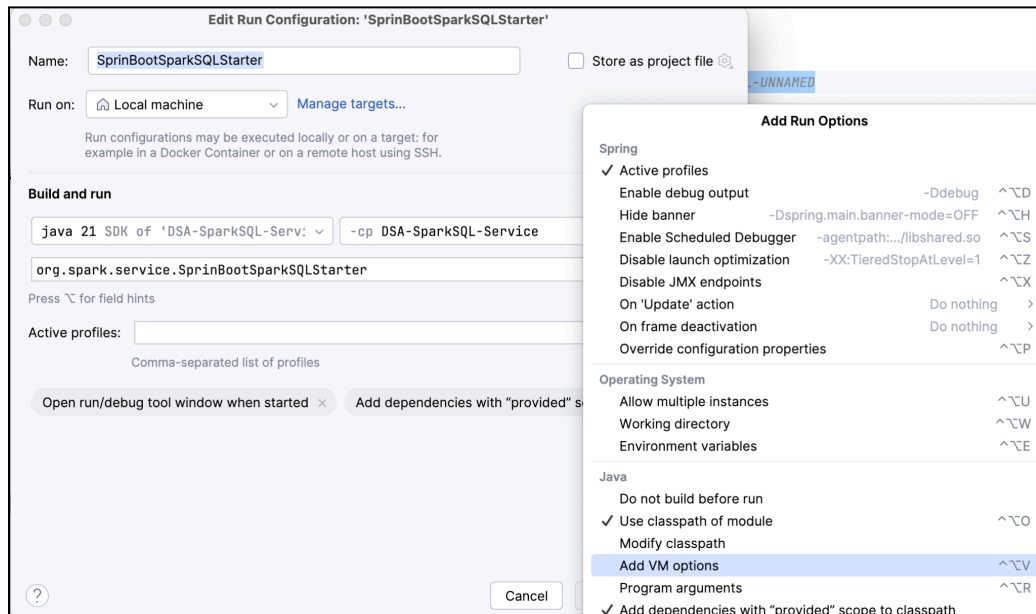
```
1 package org.spark.service;
2
3 > import ...
4
5
6
7
8
9
10 // --add-exports java.base/sun.nio.ch=ALL-UNNAMED --add-opens java.base/java.net=ALL-UNNAMED
11 @SpringBootApplication(exclude={DataSourceAutoConfiguration.class} )
12
13
14
15
16
17
18
19
20
21 }
```

A context menu is open over the class name `SprinBootSparkSQLStarter` on line 12. The menu options are:

- Run 'SprinBootSpark....main()'
- Debug 'SprinBootSpark....main()'
- Run 'SprinBootSpark....main()' with Coverage
- Profile 'SprinBootSpark....main()' with 'IntelliJ Profiler'
- Debug using HotswapAgent
- Modify Run Configuration...

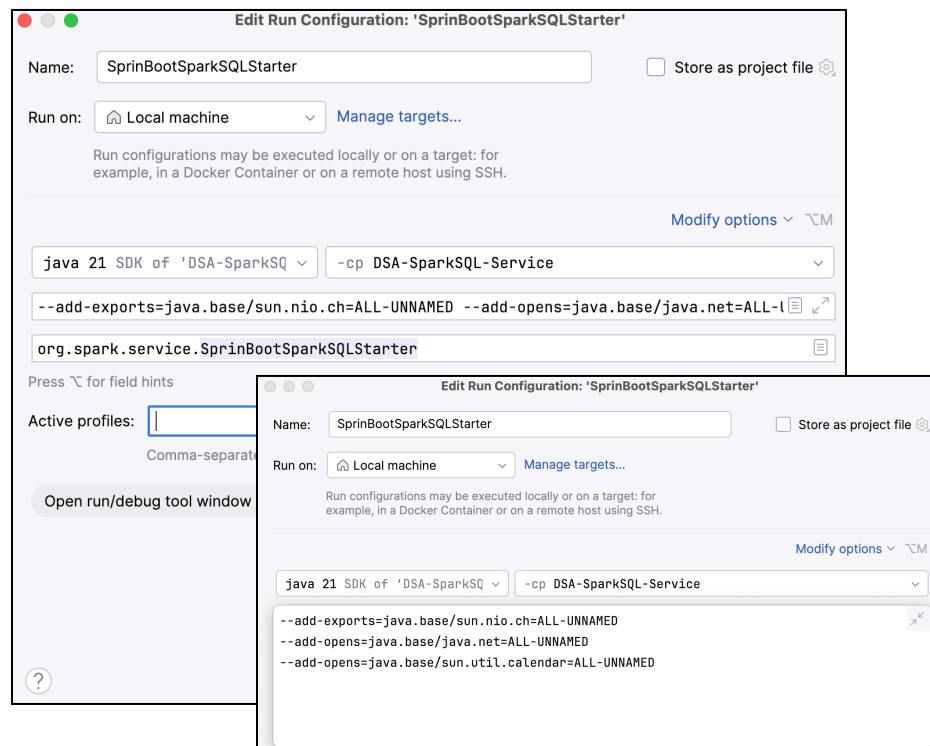
The background code shows the class `SprinBootSparkSQLStarter` extending `SpringBootServletInitializer` and implementing `SpringBootStarterService` with Spark Default Settings ... DSA".

- Click on [Modify Run Configuration] action.
- Click on [Modify Options] link:
 - Select [**Add VM Option**].

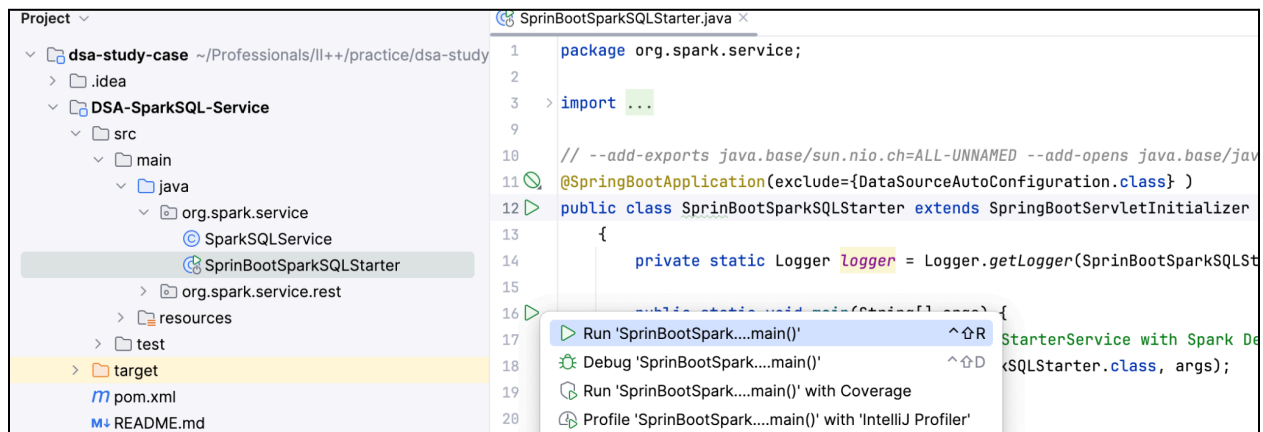


- **Copy** in [VM Options] text box:

```
--add-exports=java.base/sun.nio.ch=ALL-UNNAMED
--add-opens=java.base/java.net=ALL-UNNAMED
--add-opens=java.base/sun.util.calendar=ALL-UNNAMED
```



- **RUN** @SpringBootApplication SprinBootSparkSQLStarter



- **WORKAROUND** when running Spark SQL on **WINDOWS**: Hadoop/Winutils libs
[Links: [getorchestra](https://getorchestra.com)]:
 - Download the [winutils.exe] and [hadoop.dll] files from this Git repository:
<https://github.com/steveloughran/winutils/tree/master/hadoop-3.0.0/bin>]
 - Save [winutils.exe] and [hadoop.dll] files to a local path, e.g., D:\Hadoop\bin
 - Define the HADOOP_HOME environment variable on Windows, pointing to D:\Hadoop local path.
 - Update PATH environment variable on Windows, adding %HADOOP_HOME%\bin directory.